Pharmaceutical Pollution:

The Hudson River



By Hana Bauta

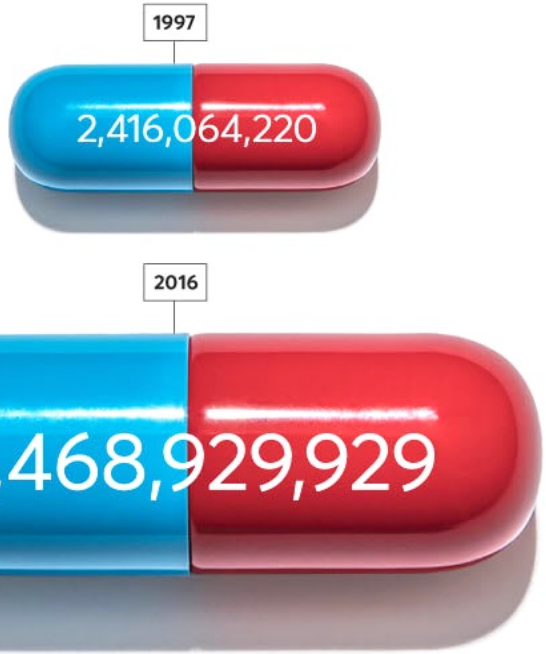
The Problem

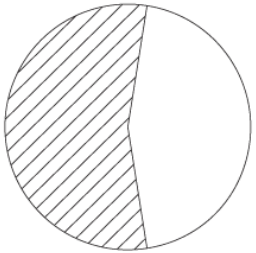
The Issue of Overmedication

Total Number of Prescriptions filled by Americans over two decades:

Fills Increased by 85% From 1997 to 2016

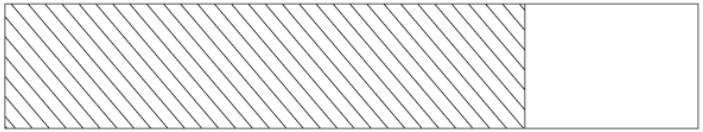
Population only increased by 21%





55%

of Americans regularly take prescription medication—four, on average.



75%

of them also take at least one over-the-counter drug regularly.

The Pharmaceutical Life Cycle

Production

Factory production of pharmaceuticals

01

Distribution

Hospitals, pharmacies and retailers to patients

02

Improper Disposal

Improper disposal (flushing or tossing) of expired/unused medication

03

Sewage Treatment

Wastewater sent to treatment facilities

04

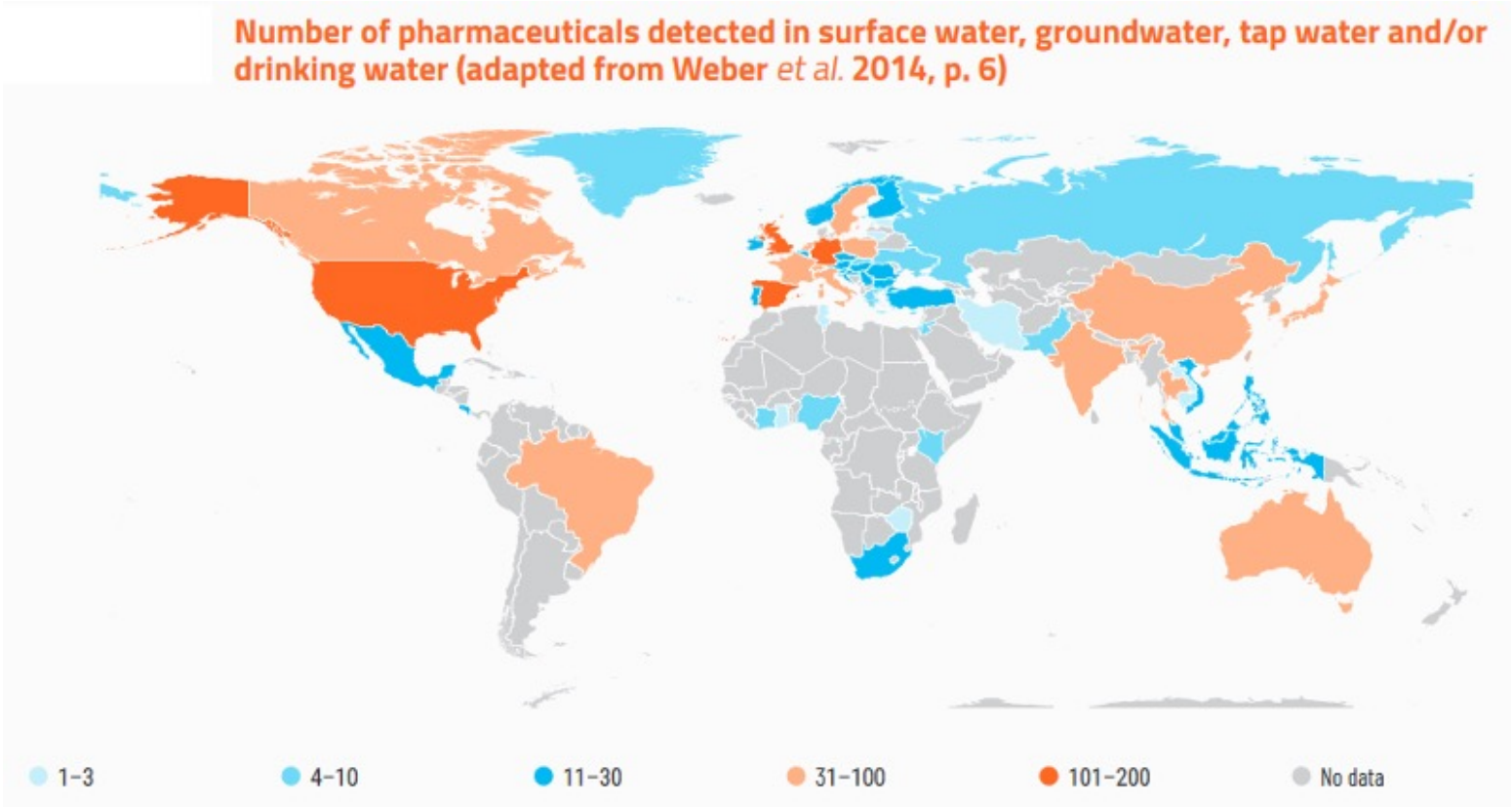
Waterways

Residual pharmaceutical compounds reenter vital water sources

05



The Pollution of Waterways



The Hudson River





The Hudson River

Pharmaceuticals in the Water

16 highly prescribed pharmaceutical compounds were detected near sewage outfalls along 250 km of the Hudson River

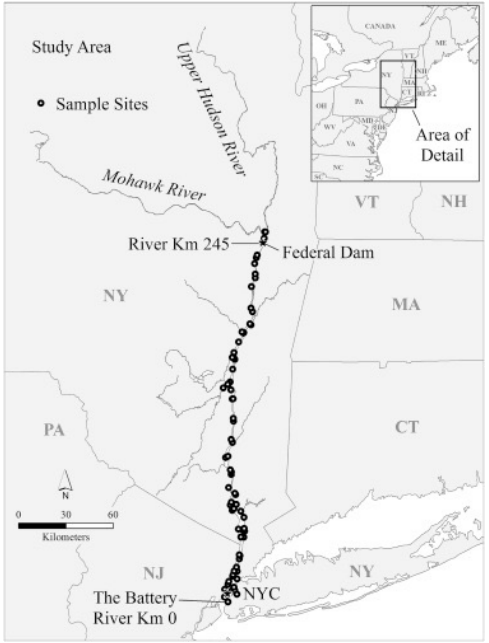
Detected Drugs

- Anti-hypertensive
- Antibiotic
- Analgesic/ Painreliever
- Anticonvulsant
- Diuretic
- Antilipemic
- Anti-ulcerative



16

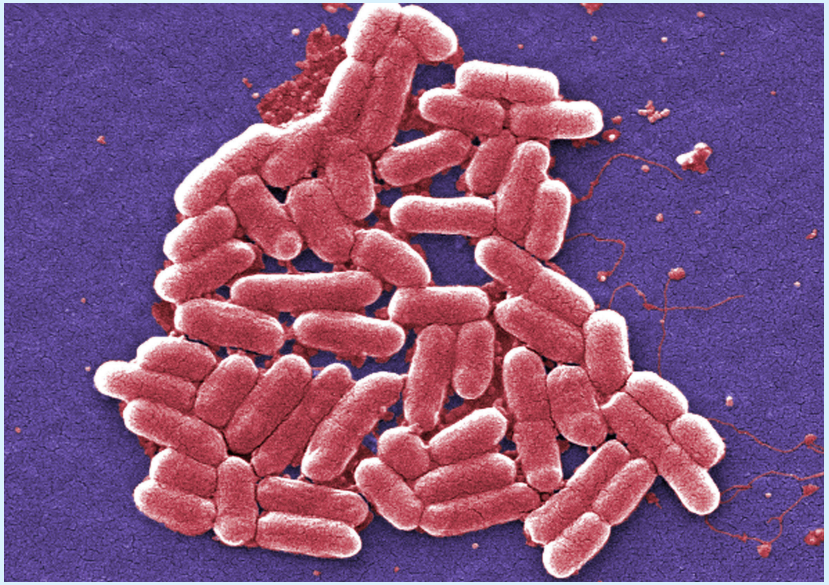
Hudson Water Samples



Environmental Consequences

The Creation of Superbugs

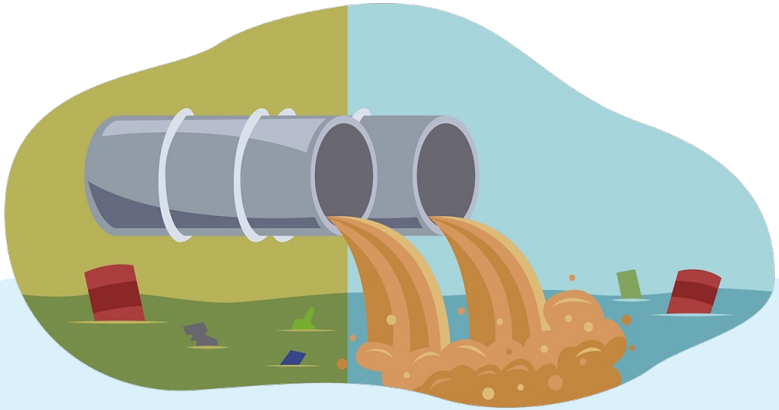
- Antibiotics in water can alter the characterization of microbes and bacteria creating drug-resistant '**superbugs**.'
- **High levels** of antibiotic-resistant bacteria in water in Hudson River Estuaries.
- WHO: "Antimicrobial resistance is one of the **top 10 global public health threats** facing humanity."



Environmental Consequences

Danger to Aquatic Ecosystems

- A Swedish study discovered human concentration levels of **hormonal drugs** among sewage exposed fish causing reduced fertility levels.
- The Cary Institute for Ecosystem Studies in NY found crayfish exposed to **antidepressants** increased levels of aggression and boldness in exposed aquatic species.



- Alterations to food web dynamics, reduced growth, and organ/hormonal changes

The Population



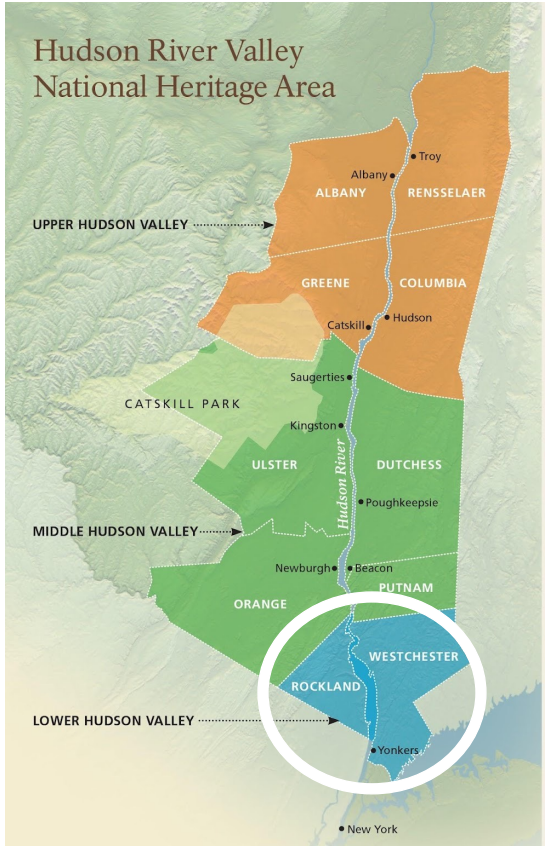
The Hudson Valley

Population at Risk

The Lower Hudson Valley region has alarming levels of pharmaceutical compounds in the water

Compound Levels

Rockland County and Westchester County tested positive for 90% of the 16 pharma compounds aforementioned



Westchester County

Total Population: 967,506

Housing Units: 353,485

Rockland County

Total Population: 325,789

Housing Units: 101,167

Total Households: 454,652

The Plan



Drug Take-Back Programs

- New York State senate passed the ’Drug Take Back Act’
- The establishment of an industry-funded, statewide pharmaceutical drug take-back program





- This includes the establishment of drug take-back collection boxes at pharmacies, police stations and mail-in options.
- This program will help protect New York State’s waterways by preventing drugs from being improperly disposed of by flushing.

Raising Awareness



- In NY **less than ⅓** of patients received information related to proper medication disposal
- Environmental health promotion
- Rockland County and Westchester County
- The promotion of **Drug Take-back Programs** and awareness of the hazards of improper drug

Educational Campaign



- Pharmaceutical education will be administered to the public through a **media campaign**
- Informational **infographic brochures**
- Distributed to each residential household with permission from the county municipalities
- Brochures will include a hashtag and website address

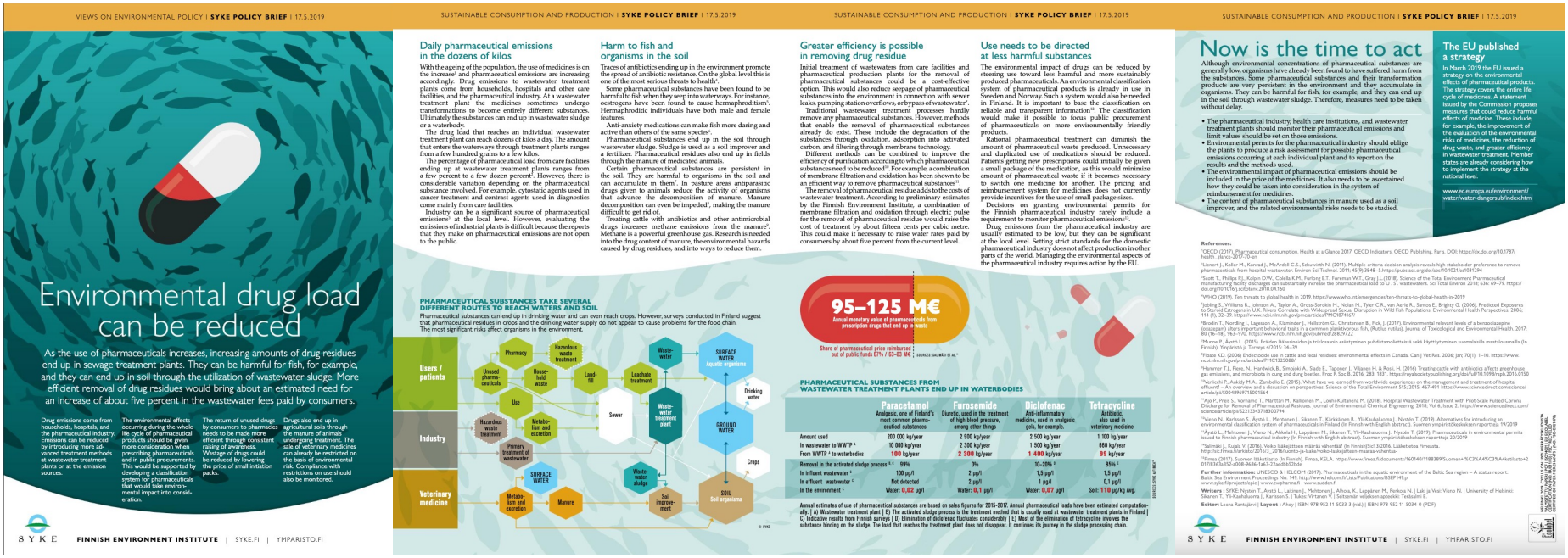
Medication Education

www.MedicationEducation.com

Paradigms



Campaign Framework



- Finnish Environment Institute

Model of Success

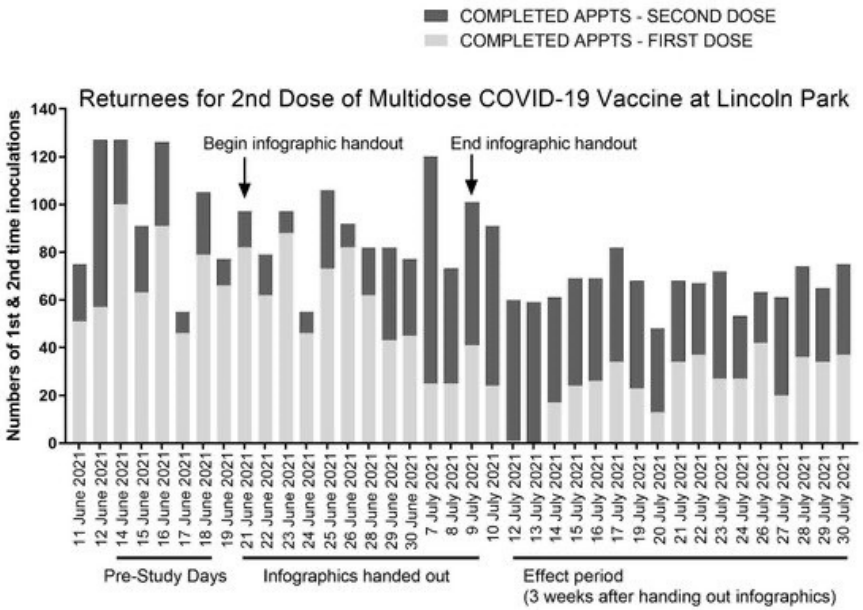
CWPharma Project

- Interreg Baltic Sea Region project
- Public awareness campaigning in Germany
- **“Don't give nature the rest. Dispose of medications properly.”**
- Campaigns conducted annually over five years



Model of Success

- Covid-19 Infographic Health Promotion Campaign
- Lincoln Park Vaccination Site in Los Angeles, CA
- Increased second-dose vaccination rates





Price



Budget Overview

Estimated Total: $92,000

Graphic Designer

$100 or more per design

Computer Programmer

Between **$1000** and **$5000** per website

$50 per month for maintenance

Printing

$0.04 per pamphlet

Around **$50** per 1000 printed brochure leaflets

Website Domain

MedicationEducation.com
$4,295

Labor & Transportation

Leaflet distributors paid per 1000 flyers for **$130** or **$15** per hour

Volunteer environmentalists

Costs

9%

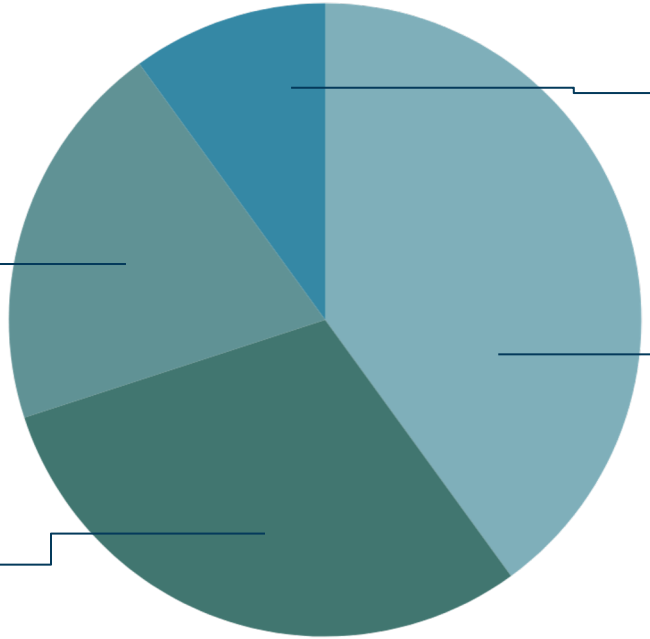
Website Development

$8,000

1%

Graphic Design

$1000



25%

Printing

$23,000

65%

Distribution

$60,000

Why Take Action?



Conclusion



The Hudson River: An Urgent Problem

- As of 2019 the Hudson River is the **second most endangered river** in the United States.
- Second largest estuary on the East Coast
- Provides habitat for various **endangered** species
- The Hudson River serves one of the **most densely populated** areas in the country.

Thank You For Your Time

Do you have any questions?



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